

COURSE TITLE : WOOD PROCESSING
COURSE CODE : 3123
COURSE CATEGORY : B
PERIODS PER WEEK : 5
SEMESTER : 3
PERIODS PER SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
I	Forest & Forestry, Forest Operations	10
	Harvesting of Timber	8
	Test	1
II	Felling	5
	Classification of trees	5
	Utilisation of small logs	4
	Identification of Timbers	4
	Test	1
III	Biological deterioration of Timber	8
	Wood Preservation	10
	Test	1
IV	Wood seasoning	17
	Test	1
Total		75

Course outcome :

Module	G.O	Student will be able to:
1	1	Understand The Role of Forest, Importance of Forestry And Utilisation of Forest Products In Wood and Paper Industry
	2	Understand the Harvesting of Timber
2	1	Understand the Felling and Transportation of Logs
	2	Comprehend the Defects in Timber
3	1	Understand the Causes and Remedies of Attack by Biological Agencies
		Understand the Wood Preservation
4	1	Understand the Seasoning of Wood

Specific outcome

MODULE I

1.1.0 Understand The Role Of Forest, Importance Of Forestry And Utilisation Of Forest Products In Wood And Paper Industry

- 1.1.1 State the forest, forestry, forest products tree
- 1.1.2 Describe types of forest and their distribution in India – tropical wet ever green forest, semi ever green forest, moist deciduous forest dry deciduous forest, dry ever green forest
- 1.1.3 Explain the role of forest – protective role, productive role and bio – aesthetic role
- 1.1.4 State social forestry
- 1.1.5 Explain the forest operations
- 1.1.6 State the working plan
- 1.1.7 Describe the organization structure of forest management
- 1.1.8 Explain the silvicultural systems
- 1.1.9 Describe plantations ,reforestation
- 1.1.10 Describe the role of forest for wood industry and paper industry
- 1.2.0 Understand the Harvesting of Timber**

- 1.2.1 State the term logging
- 1.2.3 List Logging methods
- 1.2.4 Give examples of Logging tools
- 1.2.5 Describe the logging practices in India
- 1.2.6 Explain logging waste

MODULE II

2.1.0 Understand the Felling and Transportation of Logs

- 2.1.1 State the term felling of Timber
- 2.1.2 Describe the principles of felling
- 2.1.3 Explain the methods of felling
- 2.1.4 Describe the method of storage of logs
- 2.1.5 Estimate of volume of log by Hoppus method
- 2.1.6 Compare on road and off road transportation
- 2.1.7 Describe the classification of trees
- 2.1.8 Distinguish between exogenous and endogenous trees
- 2.1.9 Illustrate the cross section of exogenous tree
- 2.1.10 State the characteristics and uses of teak, Rose wood, Aini,, Jack wood, Padauk, Mahagony, Laurel, Pali, Irul, white cedar, Eucalyptus, Pine, Chikkarassy
- 2.1.11 Describe the small logs
- 2.1.12 Explain identification of timbers by using dichotomous keys and perforated card key

2.2.0 Comprehend the Defects in Timber

- 2.2.1 Define defects in timber
- 2.2.2 Explain natural defects like knots, cross grain, splits, checks, rind galls, and draw

- figures
- 2.2.3 Explain reaction wood
- 2.24 Explain tension wood
- 2.2.4 Define figures in wood

MODULE III

3.1.0 Understand the Causes and Remedies of Attack by Biological Agencies

- 3.1.1 Understand the biological deterioration of wood
- 3.1.2 State the biological deterioration of wood
- 3.1.3 State the brown rot, white rot, soft rot, mould, sap stain
- 3.1.4 Describe the deterioration due to brown rot, white rot, soft rot
- 3.1.5 Explain the physiological requirements of wood rotting fungus wood inhibiting fungus
- 3.1.6 List wood boring insects – white ant, marine borers, powder post beetles carpenter ants and their deteriorative action on wood
- 3.1.7 List the mechanical properties of wood.

3.2.0 Understand the Wood Preservation

- 3.2.1 State the need of wood preservation
- 3.2.2 List the objectives of wood preservation
- 3.2.3 List the requirements of a good preservative
- 3.2.4 Describe oil type organic solvent type water soluble type, leachable type fixed type fire retardant type anti stain type preservatives coal tar creosote pentachlorophenol, ASCU, C.C.A, C.C.B, Zinc Chloride, zinc meta arsenite, boric acid borax powder sodium penta chloro phenate
- 3.2.5 List the types of preservation methods
- 3.2.6 List non pressure process such as brushing spraying, dipping, steeping, Osmos, process hot and cold bath, diffusion process, boucherie and modified boucherie process
- 3.2.7 Describe the above methods
- 3.2.8 List the pressure processes – full cell process, Empty cell Process
- 3.2.9 Describe full cell process and Empty cell Process
- 3.2.10 State the need of quality control in wood preservation

MODULE IV

4.1.0 Understand the Seasoning of Wood

- 4.1.1 State seasoning of wood
- 4.1.2 List the advantages of seasoned wood
- 4.1.3 Define – fiber saturation point, equilibrium moisture content, free moisture and bound Moisture
- 4.1.4 Describe the classification of timber
- 4.1.5 Describe stacking methods with figure
- 4.1.6 Describe the seasoning methods

- 4.1.7 Describe the shed type I, II, and III
- 4.1.8 Describe natural seasoning, electrical seasoning, chemical seasoning, water seasoning, solar Seasoning
- 4.1.9 State the kiln seasoning and types of Kiln
- 4.1.10 Explain kiln Seasoning
- 4.1.11 Calculate the moisture content of wood by oven dry method
Explain the defects –twist
- 4.1.1 Explain the defects –twist, spring, cup, bow, cross –shakes, honey combing surface checks, with figure.

CONTENT DETAILS

MODULE- I

Define the forest, forestry, forest products, trees type of forest and their distribution in India tropical wet evergreen forest, semi ever green forest, moist deciduous forest, dry evergreen forest. Distribution of bamboos and reed forests in India. Role of forest – protective role, productive role and bio-aesthetic role. Farm forestry. Forest operations – working plan – structure of forest management. Silvi cultural systems – clear felling systems – reforestation plantations - - intensive plantation and trench mound plantations. Logging – logging practices in India, logging waste how to avoid it. Implements used in felling and conversion.

MODULE – II

Felling – principles of felling, rules of felling, methods of felling dry storage wet storage. Water storage calculation of log volume by Hoppus Method , Off- road and on -road transportation Classification of trees – exogenous and exogenous trees. Cross-section of exogenous tree, comparative study of hardwood and soft wood. Utilization of small log, important timbers – teak, Rose wood, Aini, Jack wood, Padauk, Mahogany, Pali, Laurel, Irul, White cedar, Eucalyptus, chikkarassy – characteristics and uses of these timbers. List the natural defects – state the causes and features of natural defects with figure - knots check, split, shakes, cross grain, discoloration, compression wood,, tension wood, pith pocket .identification of timber – dichotomous keys – perforated card key – key to important timbers and identification – identification, properties and description, character of economically important Indian timbers such as teak, rose wood, jack wood, aini, rubber wood, white pine, mango, pali, ebony, bamboo, reed, eucalyptus, mahogany, accasia, irul, champ, chikkrassey, poon, kunnivaka, ebony, casurina

MODULE – III

Biological deterioration of wood – Brown rot white rot, soft rot, mould rot sap stain physiological requirement of wood rotting fungus wood inhibiting fungi. Wood boring insects white ant marine borers powder beetles, carpenter ants and their deteriorative action on wood Mechanical properties of wood – tensile strength, impact strength, hardness, compressive strength, bending strength, shear strength screw and nail holding power. Wood preservation – define preservation – need of preservation – objects of preservation. Requirements good preservative. Types of preservatives – Oil type, water soluble type, organic solvent type. Coal tar creosote, ASCU, CCA, CCB, Zinc chloride, Zinc Meta arsenite, boric acid, borax powder, and sodium pentachlorophenate. Methods of Preservation – non-pressure process and pressure process Brushing, spraying dipping, steeping, osmos process, hot and cold bath, diffusion process, boucherie and modified boucherie process Full cell process, empty cell, process. Comparison

between pressure process and non-pressure process. Preservation of non-durable timbers like rubber wood. Quality control in the preservation of wood.

MODULE – IV

Seasonings of wood define seasoning of wood need of wood seasoning. Advantages of seasoned wood. Shrinkage and swelling of wood. Fiber saturation point. Equilibrium moisture content. Free moisture and bound moisture classification of timber with respect to seasoning purpose- refractory, non-refractory and moderately refractory timbers. End coating of logs against splitting. Timber yard – stacking method - need of timber stacking one and nine stacking method open crib and close crib fashion stacking of poles fence posts and logs Seasoning methods – natural and artificial- natural (Air) seasoning, water seasoning, chemical seasoning, solar seasoning- electrical seasoning Kiln seasoning – advantageous and disadvantageous of Kiln seasoning. Types of kiln - Progressive and compartment type. Stacking methods in kiln . Kiln schedules, wet bulb and dry bulb. State the Need of vacuum drying – calculate moisture content by oven – dry method. List the Seasoning defects – draw the defects - twist, spring, cup, bow, cross shakes, honey combing, surface checks, splitting causes and remedies of the above defects

REFERENCE

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2. Barry A. Richardson, Wood preservation - construction press London
3. A hand book of Kerala timbers – KFRI, Trichur, Kerala
4. R K Maiti,. Plant fibers -
5. Dr. S.S Negi, Forests and Forestry - National Book trust , India
6. Forest Utilization – volume I & II – F R I, Dehradun
7. Lecture notes on plywood technology volume I – IPIRI, Bangalore
8. BIS Code for practice of preservation - (IS : 401 – 1982)
9. BIS Code for practice for seasoning of timbers - (IS : 141 – 1973)
10. BIS Glossary of technical term - (IS : 707 – 1976)